Three customer questions. One inspectable prototype.

Who will buy again?

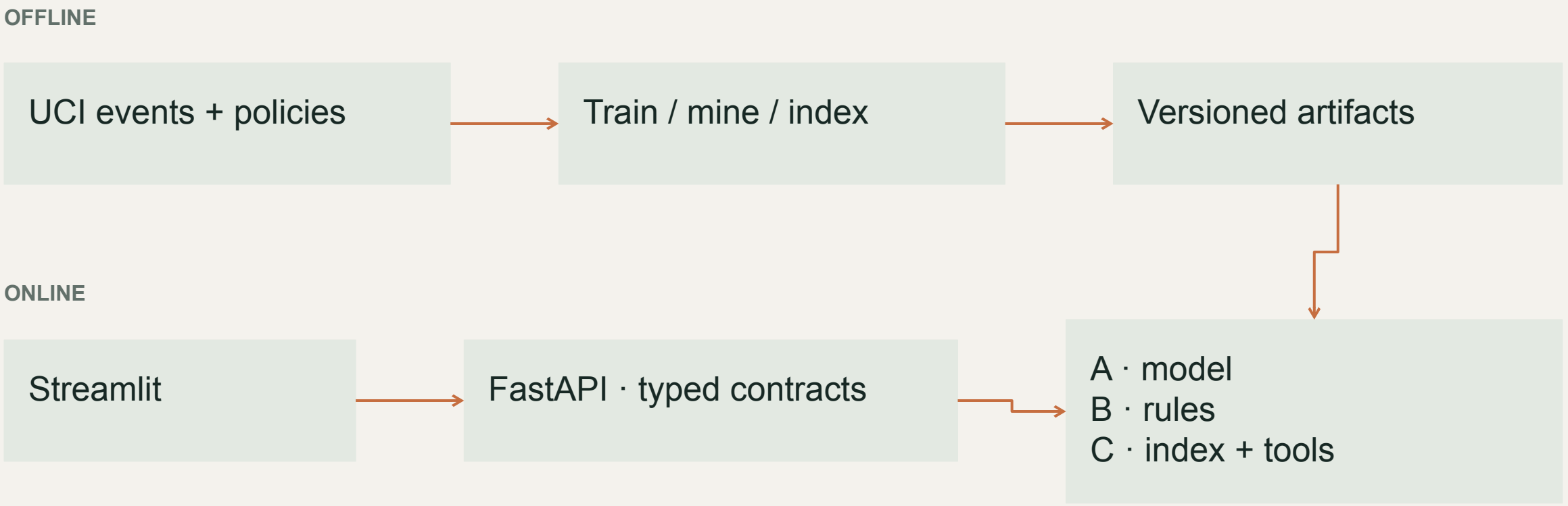
What belongs in this basket?

Which policy answers this question?

> Evidence-led decisions, with the assumptions visible.

SYSTEM DESIGN

Offline artifacts support three online paths



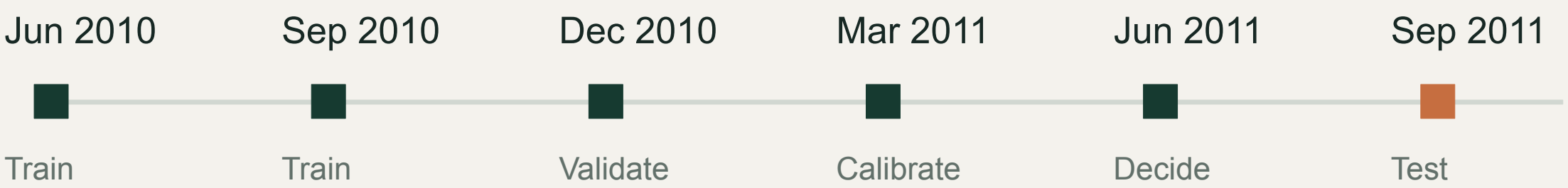
Prediction logs → PostgreSQL → operational and drift views

Source: ARCHITECTURE.md; api/lifespan.py; frontend/streamlit_app.py

Six cutoffs across five chronological stages

25 engineered features

12 serving inputs



History before each cutoff → purchase label in the next 90 days

UK giftware · GBP retained · repeat customers across periods · no future-order features

Source: retention_metrics.json → splits; src/features.py; UCI Online Retail II

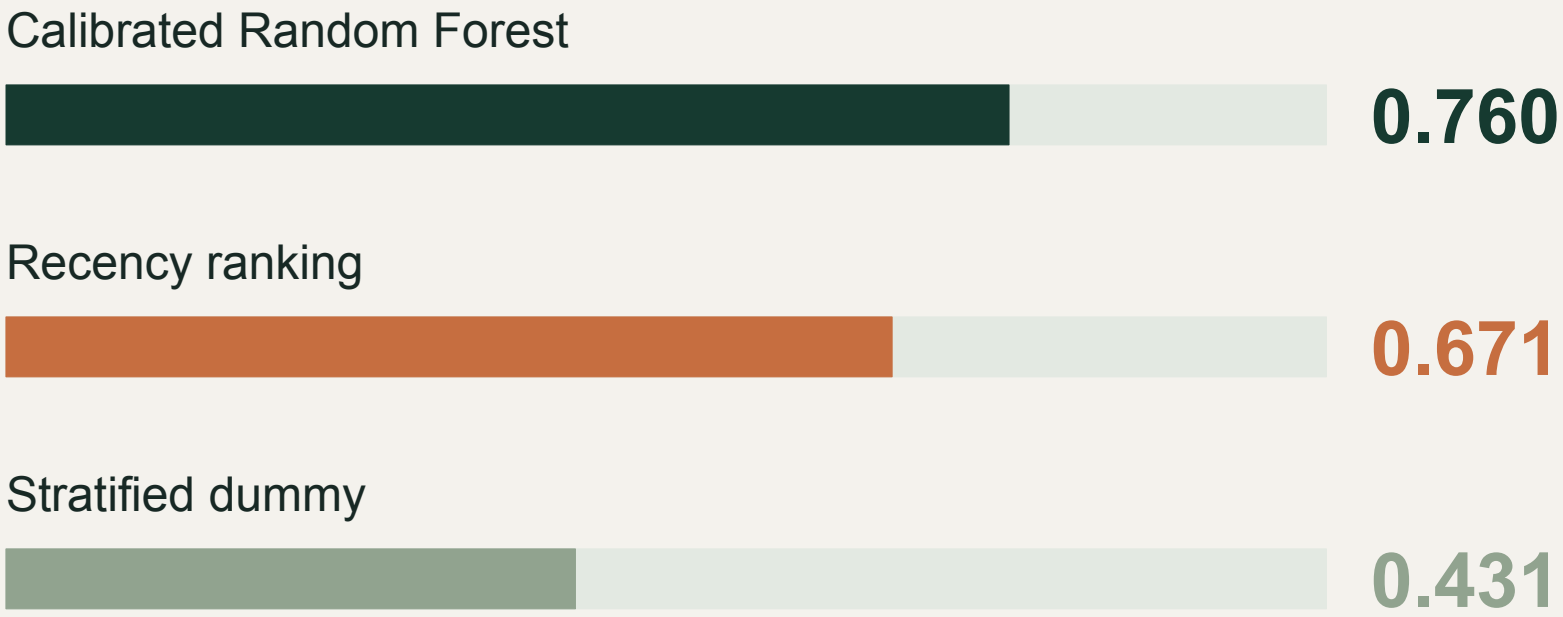
Calibration fitting and selection use different periods

Stage	Cutoff(s)	Rows
Train	Jun / Sep 2010	6,004
Validate	Dec 2010	4,249
Fit calibration	Mar 2011	4,522
Choose mapping / threshold	Jun 2011	4,918
Evaluate	Sep 2011	5,233

Estimator refit: train + validation = 10,253 snapshot rows. Reused test period disclosed.

Source: retention_metrics.json → splits; src/train.py

The clearest gain is in ranking returners



AVERAGE PRECISION

0.742–0.775

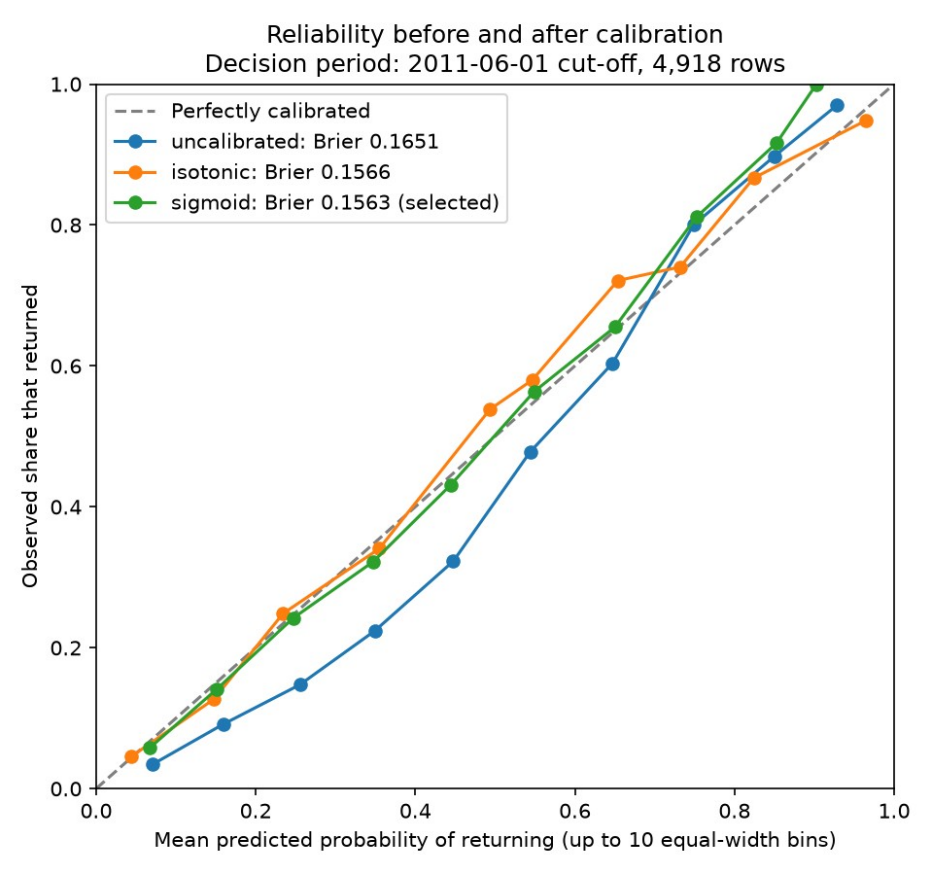
95% customer-bootstrap interval

ROC-AUC 0.786
Return prevalence 42.7%

AP difference vs recency: +0.071 to +0.105 (paired 95% interval)

Source: retention_diagnostics.json; retention_metrics.json · September 2011, N=5,233

Calibration is selected before the test period



JUNE BRIER · LOWER IS BETTER

Raw	0.16506
Isotonic	0.15665
Sigmoid	0.15631

Sigmoid selected

Later test Brier
0.19398

Brier = mean (probability − outcome)²

Source: retention_metrics.json → calibration_decision_period; figures/reliability_curve.png

20.5% is an assumed-utility policy

Offer: pM – C
No offer: –pF

$$p \geq C / (M + F)$$

C = 5 M = 25 F = 0

20%

theoretical break-even

20.5%

June sweep optimum

Purchase probability ≠ incremental coupon effect. No coupons are dispatched.

Source: src/train.py → utility/select_threshold; frontend/coupon.py; retention_metrics.json

The binary gain over recency is small

Actual / predicted	Not selected	Selected
Did not return	TN 1,606	FP 1,391
Returned	FN 393	TP 1,843

Model recall 82.42%

Recency recall 81.93%

Equal selection count · precision 56.99% · AP interval 0.7425–0.7749

Source: retention_diagnostics.json · September test; selection count 3,234

Inputs were reduced; fairness gaps remain

25
engineered offline

− 13
removed from serving

= 12
model inputs

Five strong thin-history recall disparities remain.

Operational mitigation: human review of every proposed action.
No automatic coupon dispatch; no claim of statistical parity.

Location fields are excluded from serving; retained offline for auditing. Tiny unknown-region group is inconclusive.

Source: FEATURE_INVENTORY.md; fairness_audit.json; minimisation_audit.json; PROMOTION_POLICY.md

Performance evidence arrives after the label matures

NOW

Log score, version and features
Check volume, latency and drift

→

AFTER 90 DAYS

Join mature purchase outcomes
Check label completeness

AP / ROC / Brier + subgroup recall

Review alert → retraining PR → approval gate

Live drift checks exist. Delayed-label backfill and external alert delivery are not implemented.

Source: docs/MONITORING.md; prediction_log; /metrics · delayed-label flow is a documented design

White holder → red holder

A · white hanging heart t-light holder → B · red hanging heart t-light holder

23.1%

confidence · P(B | A)

4.70×

lift · relative to P(B)

3.54%

support · P(A and B)

Past co-purchase evidence, not a causal or personal-preference claim.

Source: recommendation rule export; frontend/basket.py → learned rule explorer

A rule must clear three different tests

Support ≥ 0.5%

P(A and B) ≥ 129 / 25,601 baskets

Confidence ≥ 10%

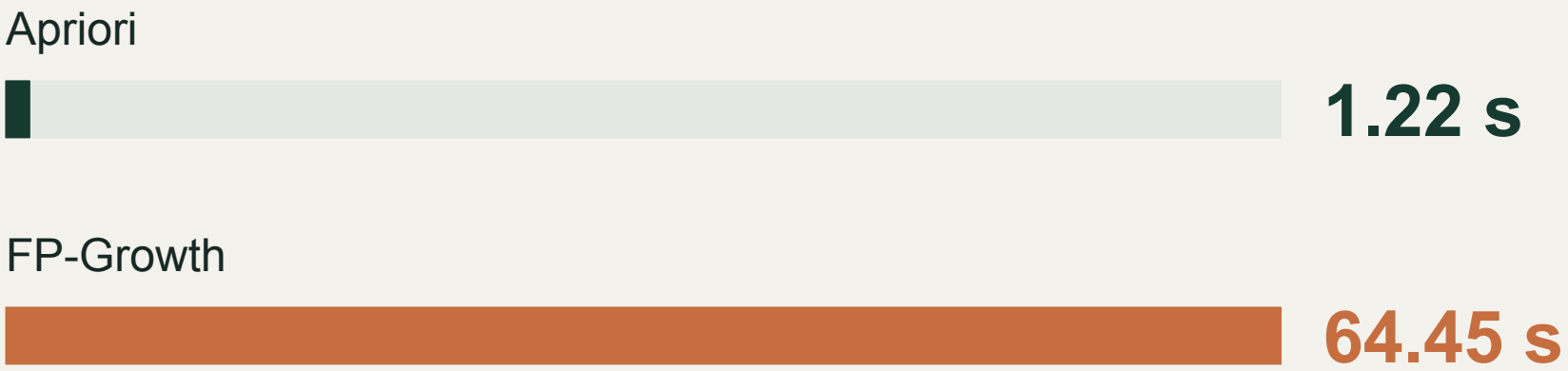
P(B | A) = P(A and B) / P(A)

Lift ≥ 1.2

P(B | A) / P(B) · combined itemset ≤ 3 products

Source: recommendation_metrics.json → header; defence_facts.json → q12_lift_filter

FP-Growth serves; Apriori was faster to mine here



SAME 3,567 RULE KEYS

2.71 MB

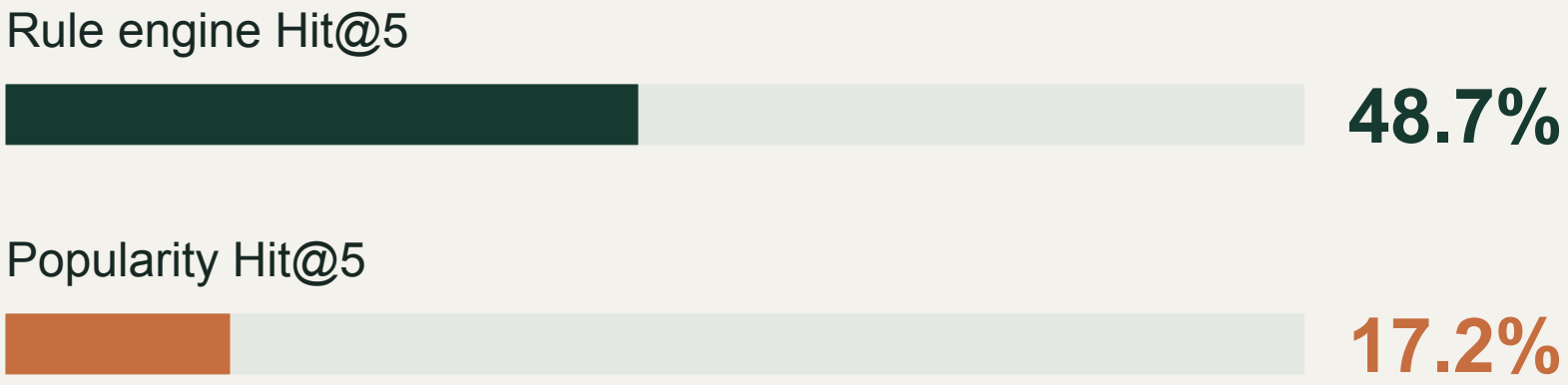
sparse matrix
119.99 MB dense equivalent

0.222 ms

local serving p95
indexed lookup; excludes HTTP/UI

Source: recommendation_metrics.json → header.timing / memory_report; rules_scaling.json

Better recovery, but a narrow catalogue



83.7%

baskets fire a rule

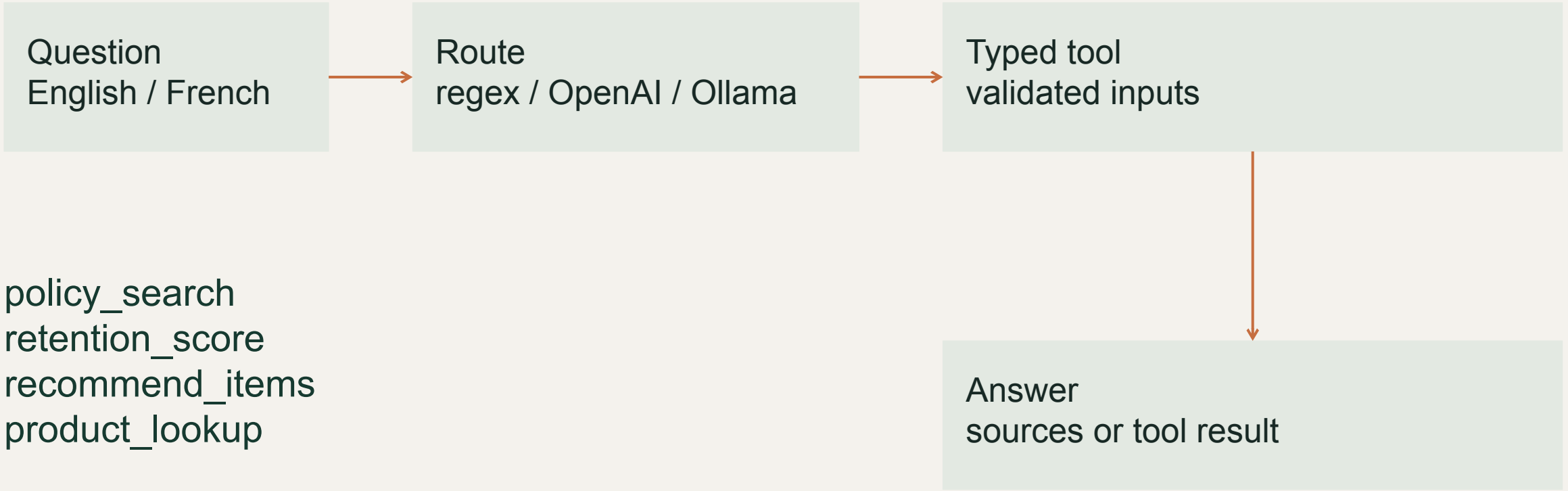
7.7%

catalogue products suggested

Half-basket masking; a hit means ≥1 hidden item among 5 suggestions. 400 single-item baskets excluded.

Source: recommendation_metrics.json → final_test · 5,884 eligible baskets

The assistant routes before it answers



policy_search
retention_score
recommend_items
product_lookup

Low relevance → refuse + fictional support handoff

Source: api/routers/chat.py; src/rag/providers.py; src/rag/index.py

Retrieval similarity is a routing heuristic

Similarity	Policy-answer behaviour
Below 0.25	Refuse + fictional bilingual support handoff
0.25 to below 0.45	Answer with caution and retrieved citations
0.45 and above	Normal grounded-answer path

Similarity is not the probability of truth.

Citations and French language match still need factual answer review.

Source: src/rag/guardrails.py; api/routers/chat.py; tests/test_passage_confidence.py

Correct routing does not ensure correct prose

Provider	Tool routing / 40	Correct answers / 18
Regex	97.5%	Extractive baseline
OpenAI	95.0%	11
Ollama	92.5%	3

8 / 8 out-of-scope questions refused by each provider.

Development questions; written answers scored by one AI rater. Independent bilingual review remains.

Source: assistant_*_metrics.json; answer_correctness.json · historical development evaluations

Local evidence is strong; cloud evidence is pending

Verified locally	Evidence
Tests	428 passed · 90% coverage
API + Streamlit	16 HTTP + 7 interaction checks
Containers	Rebuilt local stack · readiness checked

GitHub / AWS code ready; sign-in and execution pending

Cloud evidence and public release remain pending authentication.

Source: logs/completion_*; completion_coverage_summary.txt; CAPSTONE_REQUIREMENTS_STATUS.md

Three heads, three kinds of evidence

1

Head A · purchase probability

Better ranking; coupon effect remains unmeasured

2

Head B · basket associations

Better hidden-item recovery; limited catalogue reach

3

Head C · grounded tool use

Inspectable sources; answer correctness needs review

Source: MODEL_CARD.md; CAPSTONE_REQUIREMENTS_STATUS.md; AI_USE.md

Every headline maps to a saved source

A · model + uncertainty	retention_model_metadata.json retention_metrics.json · retention_diagnostics.json
B · rules + scaling	recommendation_metrics.json rules_scaling.json · defence_facts.json
C · routing + correctness	assistant_*_metrics.json answer_correctness.json · tests/golden/
Delivery + governance	logs/completion_* · fairness/minimisation audits CAPSTONE_REQUIREMENTS_STATUS.md